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Harrier Engine Service Manual

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1 Introduction and Safety

This manual covers diagnosis and service of the Harrier engine, including timing, position sensing, lubrication, forced induction, and mounting sub-systems. Allow the engine to cool fully before draining fluids or removing turbocharger components to avoid burns. Disconnect the battery negative terminal before servicing position sensors or ignition components. Read each procedure completely and observe all torque sequences before beginning work.

2 Engine System Overview

The Harrier engine integrates the timing belt TB-3320 with idler TB-3322, the timing chain TC-3340, the variable valve timing solenoid VVT-3360, the crankshaft position sensor CPS-3380, and the camshaft position sensor CMP-3382. Lubrication and ventilation are handled by the valve cover gasket VCG-3401, the oil filter kit OFK-3420, and the PCV valve PCV-3490, while the engine mount ENM-3450 and turbocharger TBO-3470 complete the assembly. These components share platform conventions with the Falcon Engine Service Manual . Identify the affected sub-system before beginning diagnosis.

3 Diagnosis

Begin engine diagnosis by retrieving misfire codes, which include the general P0300 and the cylinder-specific P0301, P0302, P0303, and P0304. Confirm fuel, spark, and compression before condemning any single component. Record freeze-frame data and note the conditions under which the misfire occurs. Verify the customer concern before clearing any codes.

3.1 Misfire Diagnostic Trouble Code Reference

Code P0300 indicates a random or multiple-cylinder misfire, while P0301, P0302, P0303, and P0304 identify misfires isolated to cylinders one through four respectively. A pattern of cylinder-specific codes points to a localized fault, whereas P0300 alone suggests a shared cause such as fuel pressure or timing. Record all stored codes before beginning component testing. Compare freeze-frame data across codes to narrow the cause.

3.2 Misfire Diagnosis Procedure

When diagnosing P0300, P0301, P0302, P0303, or P0304, verify ignition, fuel delivery, and mechanical timing in sequence. A skewed signal from the crankshaft position sensor CPS-3380 or camshaft position sensor CMP-3382 can produce misfire codes without a true combustion fault, so confirm sensor correlation early. Inspect timing components if multiple cylinders are affected. Document compression readings before concluding the diagnosis.

4 Timing Sub-System

The timing sub-system comprises the timing belt TB-3320, idler TB-3322, timing chain TC-3340, and variable valve timing solenoid VVT-3360. Correct valve timing is essential for combustion and emissions performance. The service approach parallels the Falcon Engine Service Manual :: Timing System . Always verify timing marks before rotating the engine.

4.1 Timing Belt and Tensioner Replacement

Replace the timing belt TB-3320 and idler TB-3322 as a set at the specified interval to prevent catastrophic valve damage. Set all timing marks, then verify belt routing and tensioner deflection before final assembly. The detailed procedure matches the Falcon Engine Service Manual :: Timing Belt and Tensioner Service . Rotate the engine by hand two full turns to confirm timing before starting.

4.2 Timing Chain Service

The timing chain TC-3340 is a long-life component but may stretch or rattle on cold start when worn. Inspect guides and tensioner for wear whenever the front cover is removed. The replacement steps align with the Falcon Engine Service Manual :: Timing Chain Service . Verify timing mark alignment after installation.

5 Variable Valve Timing Solenoid Replacement

The variable valve timing solenoid VVT-3360 adjusts cam phasing in response to control module commands. Replace the VVT-3360 when phaser response is sluggish or correlation codes are set. The procedure follows the Falcon Engine Service Manual :: VVT Solenoid Replacement . Clean the solenoid screen and verify oil supply before installation.

6 Position Sensor Sub-System

The crankshaft position sensor CPS-3380 and camshaft position sensor CMP-3382 provide rotational reference signals to the engine control module. Faults in either sensor can cause no-start, stalling, or misfire codes. The diagnostic approach parallels the Falcon Engine Service Manual :: Position Sensors . Verify air gap and connector condition before replacement.

6.1 Crankshaft and Camshaft Position Sensor Replacement

Replace the crankshaft position sensor CPS-3380 following the steps in the Falcon Engine Service Manual :: Crankshaft Position Sensor Replacement . Replace the camshaft position sensor CMP-3382 following the Falcon Engine Service Manual :: Camshaft Position Sensor Replacement . Perform any required crankshaft variation relearn after installation. Verify signal correlation with a scan tool before returning the vehicle.

7 Lubrication and Ventilation Sub-System

The lubrication and ventilation sub-system includes the oil filter kit OFK-3420, the PCV valve PCV-3490, and the valve cover gasket VCG-3401. Proper oil flow and crankcase ventilation protect the engine and reduce emissions. The layout parallels the Falcon Engine Service Manual :: Lubrication and Crankcase Ventilation . Use only the specified oil grade and capacity.

7.1 Oil and Filter Service

Service the oil filter kit OFK-3420 at the specified interval, replacing the filter element and sealing gaskets each time. Drain the oil while warm and refill to the correct level. The procedure matches the Falcon Engine Service Manual :: Oil and Filter Service . Reset the oil life monitor after service.

7.2 PCV Valve Replacement

The PCV valve PCV-3490 regulates crankcase ventilation and prevents oil pull-over and sludge. Replace the PCV-3490 when rough idle, oil consumption, or a rattling valve is found. The steps follow the Falcon Engine Service Manual :: PCV Valve Replacement . Verify hose condition and routing after installation.

7.3 Valve Cover Gasket Replacement

Replace the valve cover gasket VCG-3401 when oil seepage is found at the cover sealing surface. Clean the mating surfaces thoroughly and torque the cover fasteners in sequence to prevent leaks. Inspect spark plug well seals on the VCG-3401 assembly for oil intrusion. Use new fasteners where specified.

8 Forced Induction Sub-System

The turbocharger TBO-3470 boosts intake pressure to increase power output. Inspect the TBO-3470 for shaft play, oil leakage, and wastegate operation when boost-related symptoms appear. Always check oil supply and return lines for restriction.

9 Turbocharger Replacement

Allow the engine to cool, then disconnect the oil and coolant lines before removing the TBO-3470 turbocharger. Prime the new unit with clean oil before starting to prevent bearing damage. Torque all mounting fasteners and clamps to the values in the specifications section. Verify boost pressure after installation.

10 Engine Mount Sub-System

The engine mount ENM-3450 isolates engine vibration and maintains powertrain alignment. A worn or collapsed ENM-3450 causes excessive movement and clunking under load. Inspect the mount during any major engine service.

11 Engine Mount Replacement

Support the engine with a jack before removing the ENM-3450 mount fasteners. Replace the mount, then torque the fasteners to the values listed in the specifications section. Verify clearance and alignment after installation.

12 Specifications and Torque Values

This section lists fastener torque sequences, fluid capacities, and clearance tolerances for all Harrier engine sub-systems. Always use a calibrated torque wrench and follow the specified tightening sequence. Refer to the relevant procedure section for component-specific values.